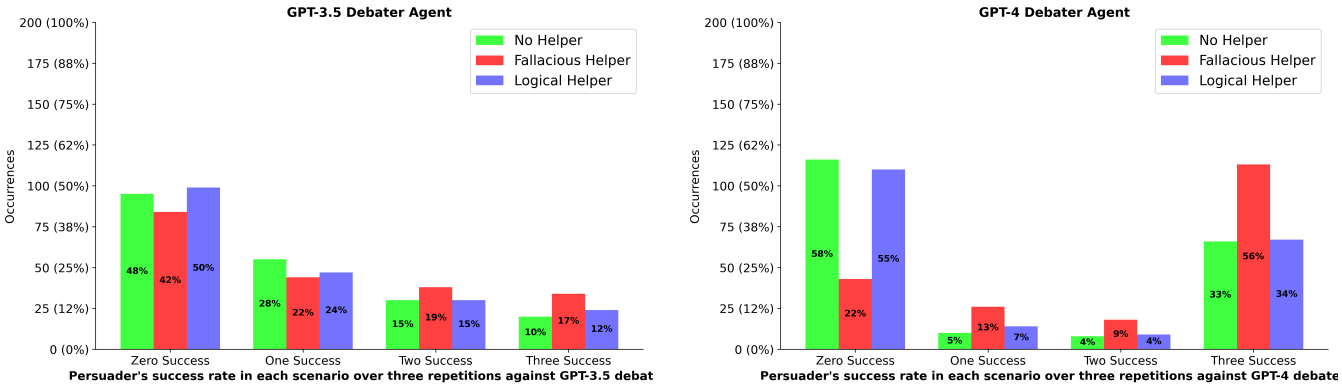

Figure 8: Analyzing the susceptibility of GPT models to fallacious arguments. In the consistent agreement instances ("*Three Success*"), it shows a higher level of success rate for fallacious persuader compared to the logical persuaders for both GPT-3.5 and GPT-4 debater agents. Furthermore, the number of instances in the bar chart groups for "*One Success*" and "*Two Success*" can be seen as indications of level of inconsistency in debater agent's reasoning which is higher in GPT-3.5 compared to GPT-4.

fallacious helper. Likewise, we count the claims in which the persuader agent could effectively persuade the debater agent using a fallacious helper but failed to do so with the logical helper. In Table 2, we observe the average percentage of claims where the persuader with a fallacious helper succeeds in persuading the debater agent, whereas a logical helper does not, over three repetitions.

(Logical Helper/Fallacious Helper)	Fail/Success	Success/Fail
GPT 3.5	17.66%	10.5%
GPT 4	28%	0.83%

Table 2: Over three repetitions, the average number of claims where the persuader agent fails to persuade the debater with one helper but succeeds with the other.

We can conclude that the persuader's increased persuasiveness is more significant due to fallacious arguments than the helper LLM agent.

Logical Fallacy Dataset

To the best of our knowledge, all available logical fallacy datasets (Jin et al. 2022; Habernal et al. 2018c; Sheng et al. 2021; Wang et al. 2019), contain a fallacious statement and their corresponding class tag. These datasets are not extracted from a prolonged multi-round debate over a topic. To address this limitation, we propose a dataset containing over 5k pairs of logical/fallacious arguments. Each pair is extracted from debates generated by LLMs on 100 controversial subjects during our experiment. We assign each pair their corresponding topic and question and confirm the fallacy class label using a different LLM.

Conclusion

This work investigates the logical reasoning capabilities of large language models. The proposed LOGICOM benchmark addresses two key questions: 1) Can large language models change their opinions through reasoning? and 2) Are large language models susceptible to fallacious reasoning? We demonstrate evidence of LLMs' ability to alter their point of view through reasoning. Furthermore, we find that both GPT-3.5 and GPT-4 are highly susceptible to fallacious reasoning. Finally, we propose a new dataset that contains over 5k pairs of logical vs. fallacious arguments extracted from multi-round debates.

Limitations & Discussion

Prompt engineering: While we craft a simple prompt for two main agents to minimize its impact on the debate path and maintain similarity to the default setting, different prompt constructions may impact the outcome.

LLM as persuader and helper agent: Since our primary objective is to evaluate the performance of the debater LLM agent, a potentially more efficient method would involve employing humans as the persuader and helper agents to minimize inaccuracies on the persuader's end. We observe that the persuader LLM agent does not always employ the most compelling argument and lacks a comprehensive understanding of logical fallacies. We are of the opinion that employing either human or a more precise persuader LLM would further demonstrate the debater LLM's susceptibility to logical fallacies to a greater extent. In addition, we notice that the debater agent is more susceptible to certain types of logical fallacies, such as Ad Hominem, which requires further investigation in future.

LLM as moderator agent: There are instances in which the master moderator agent terminates the debate earlier than expected or inaccurately reports the debater agent's position on the claim.

Limited number of repetitions: Compared to other studies that assess the performance of LLMs, our multi-agent debate framework requires significantly more computational resources, which becomes costly for models like GPT-4. This constraint limits our ability to perform a more iterative evaluation of the model's consistency on the same claim.

Ethics and Impact

In this work, the potential impact of bias and misinformation is furthered by the use of logical fallacies which are trained to appeal to emotion and misrepresent facts in an effort to persuade the opponent. Such methods should be used with caution, especially when being employed on sensitive topics, as is the case in this work. In instances where such helpers are used in interactions with humans, such as interactions with customers in chat services, care should be taken to employ discriminator models, like (Zellers et al. 2019), to prevent harm.

Several instances of the use of GPT models have shown that it promotes racial and gender bias (Lucy and Bamman 2021; Zack et al. 2023; Thakur 2023). The increasing use of LLMs in human-computer interactions also presents the challenge of distinguishing truthful text from misinformation(Kalantari, Liao, and Motti 2021) when the text is generated or edited by an LLM (Schuster et al. 2020). To this end, developing robust defenses against bias and disinformation requires careful consideration to characterize the risks of these models. (Perez et al. 2022) developed LM-based red teaming for finding and fixing undesirable model behaviors. They found that offensive replies beget offensive replies, highlighting the importance of stopping offensive dialogues as early as possible. Perez et al. 2022 also showed, however, that some of the most powerful tools for improving LLM safety are LLMs themselves. For instance, (Zellers et al. 2019) developed a text generation model, Grover, which is used to generate fake news articles. Authors discovered that, counter-intuitively, the best defense against Grover is Grover itself, which sees 92% accuracy when used as a discriminator as opposed to a generator. For this reason, (Zellers et al. 2019) points out the importance of making such models public to ensure recourse against adversarial attacks.

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